



WOLLO University

Department of software engineering

Exit multiple choice questions for

AI and big data

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1) Artificial Intelligence is about_____.

- a. Playing a game on Computer
- b. Making a machine Intelligent
- c. Programming on Machine with your Own Intelligence
- d. Putting your intelligence in Machine

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Answer: b. Making a machine Intelligent.

Explanation: Artificial Intelligence is a branch of Computer science, which aims to create intelligent machines so that machine can think intelligently in the same manner as a human does.

2) Who is known as the -Father of AI"?

- a. Fisher Ada
- b. Alan Turing
- c. John McCarthy
- d. Allen Newell

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Answer: c. John McCarthy

Explanation: John McCarthy was a pioneer in the AI field and known as the father of Artificial intelligence. He was not only the known as the father of AI but also invented the term Artificial Intelligence.

3) Select the most appropriate situation for that a blind search can be used.

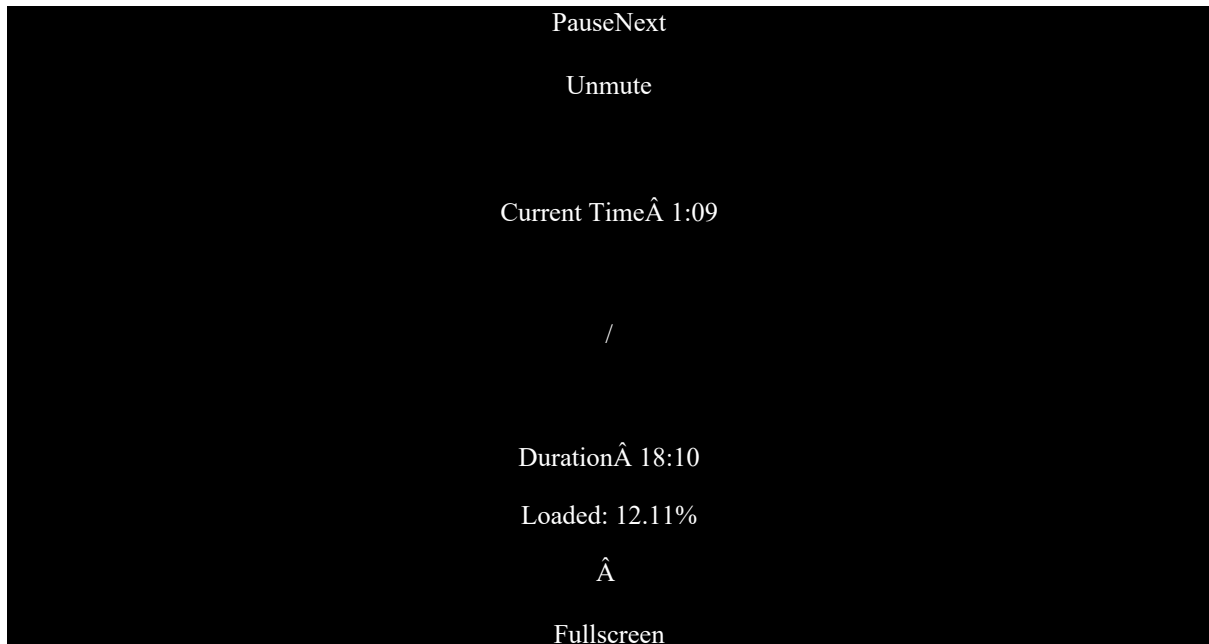
- a. Real-life situation
- b. Small Search Space
- c. Complex game
- d. All of the above

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Answer: b. Small Search Space

Explanation: Blind Search is also known as uninformed search, and it does not contain any domain information such as closeness, location of the goal, etc. Hence the most appropriate situation that can be used for the blind search is Small-search Space.

4) The application/applications of Artificial Intelligence is/are



- a. Expert Systems
- b. Gaming
- c. Vision Systems
- d. All of the above

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Answer: d. All of the above

Explanation: All the given options are the applications of AI.

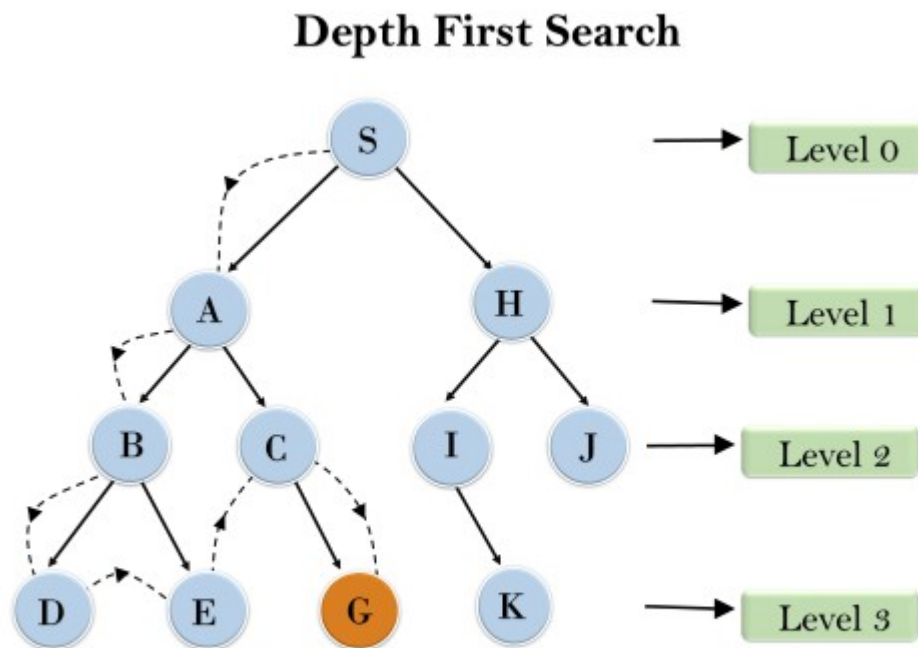
5) Among the given options, which search algorithm requires less memory?

- a. Optimal Search
- b. Depth First Search
- c. Breadth-First Search
- d. Linear Search

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Answer: b. Depth First Search

Explanation: The Depth Search Algorithm or DFS requires very little memory as it only stores the stack of nodes from the root node to the current node.



6) If a robot is able to change its own trajectory as per the external conditions, then the robot is considered as the__

- a. Mobile
- b. Non-Servo
- c. Open Loop
- d. Intelligent

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Answer: d. Intelligent

Explanation: If a robot is able to change its own trajectory as per the external conditions, then the robot is considered intelligent. Such type of agents come under the category of AI agents or Rational Agents.

7) Which of the given language is not commonly used for AI?

- a. LISP

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- b. PROLOG
- c. Python
- d. Perl

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Answer: d. Perl

Explanation: Among the given languages, Perl is not commonly used for AI. LISP and PROLOG are the two languages that have been broadly used for AI innovation, and the most preferred language is Python for AI and Machine learning.

8) A technique that was developed to determine whether a machine could or could not demonstrate the artificial intelligence known as the____

- a. Boolean Algebra
- b. Turing Test
- c. Logarithm
- d. Algorithm

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Answer: b. Turing Test

Explanation: In the year **1950**, mathematician and computing pioneer **Alan Turing** introduced a test to determine whether a machine can think like a human or not, which means it can demonstrate intelligence, known as the **Turing Test**. It was based on the "**Imitation game**" with some modifications. This technique is still a measure of various successful AI projects, with some updates.

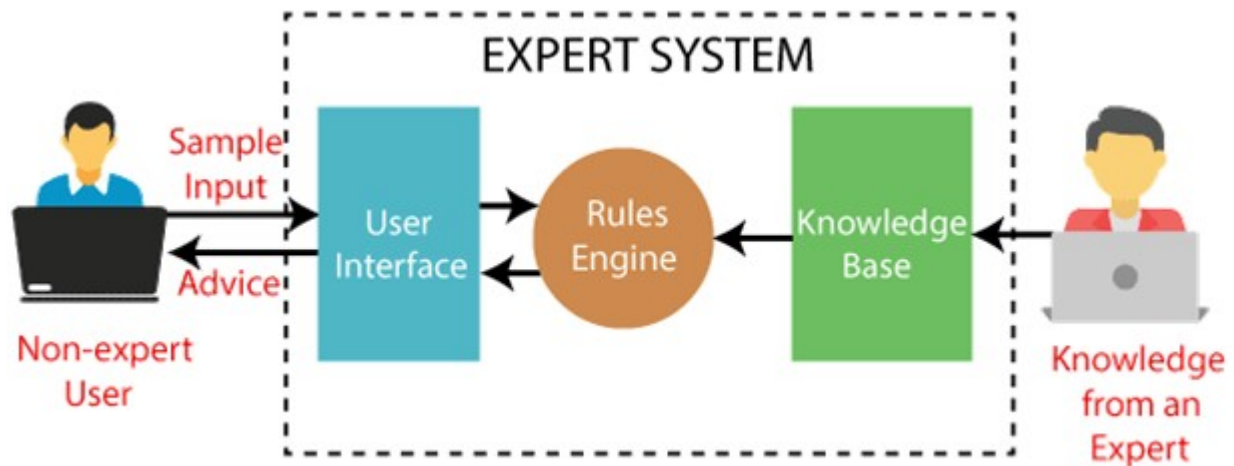
9) The component of an Expert system is_____.

- a. Knowledge Base
- b. Inference Engine
- c. User Interface
- d. All of the above

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Answer: d. All of the above

Explanation: Expert system is a part of AI and a computer program that is used to solve complex problems, and to give the decision-making ability like human. It does this with the help of a **Knowledge base, Inference engine, and User interface**, and all these are the components of an Expert System.



10) Which algorithm is used in the Game tree to make decisions of Win/Lose?

- a. Heuristic Search Algorithm
- b. DFS/BFS algorithm
- c. Greedy Search Algorithm
- d. Min/Max algorithm

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Answer: d. Min/Max Algorithm

Explanation: A game tree is a directed graph whose nodes represent the positions in Game and edges represent the moves. To make any decision, the game tree uses the Min/Max algorithm. The Min/Max algorithm is the preferred one over other search algorithms, as it provides the best move to the player, assuming that the opponent is also playing Optimally.

11) The available ways to solve a problem of state-space-search.

- a. 1
- b. 2
- c. 3
- d. 4

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Answer: b. 2

Explanation: There are only two ways to solve the problems of state-space search.

12) Among the given options, which is not the required property of Knowledge representation?

- a. Inferential Efficiency
- b. Inferential Adequacy
- c. Representational Verification
- d. Representational Adequacy

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Answer: C. Representational Verification

Explanation: Knowledge representation is the part of Artificial Intelligence that deals with AI agent thinking and how their thinking affects the intelligent behavior of agents. A good knowledge representation requires the following properties:

- **Representational Accuracy**
- **Inferential Adequacy**
- **Inferential Efficiency**
- **Acquisitional efficiency**

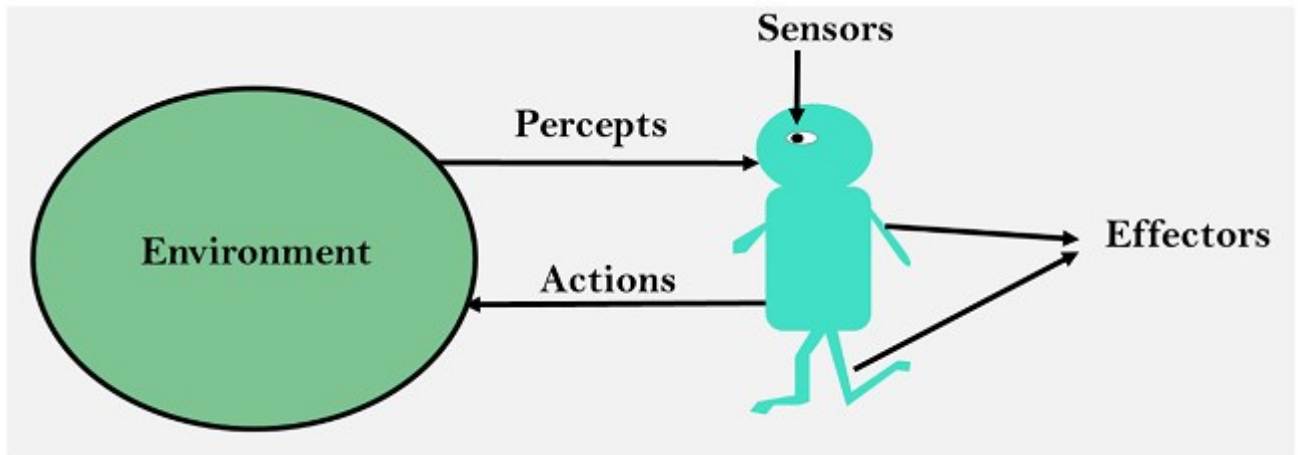
13) An AI agent perceives and acts upon the environment using_____.

- a. Sensors
- b. Perceiver
- c. Actuators
- d. Both a and c

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Answer: d. Both a and c.

Explanation: An AI agent perceives and acts upon the environment using Sensors and Actuators. With Sensors, it senses the surrounding, and with Actuators, it acts on it.



14) Which rule is applied for the Simple reflex agent?

- a. Simple-action rule
- b. Simple &Condition-action rule
- c. Condition-action rule
- d. None of the above

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Answer: c. Condition-action rule

Explanation: The simple reflex agent takes decisions only on the current condition and acts accordingly; it ignores the rest of history; hence it follows the Condition-action rule.

15) Which agent deals with the happy and unhappy state?

- a. Utility-based agent
- b. Model-based agent
- c. Goal-based Agent
- d. Learning Agent

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Answer: a. Utility-based agent

Explanation: Utility-based agent uses an extra component of utility that provides a measure of success at a given state. It decides that how efficient that state to achieve the goal, which specifies the happiness of the agent.

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16) Rational agent always does the right things.

- a. True
- b. False

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Answer: a. True

Explanation: Rational agent has clear preference, goal, and acts in a way to maximize its performance. It is said that it always does the right things, which means it gives the best performance for each action.

17) Which term describes the common-sense of the judgmental part of problem-solving?

- a. Values-based
- b. Critical
- c. Analytical
- d. Heuristic

Hide Answer Workspace

Answer: d. Heuristic

Explanation: In problem-solving, the Heuristic describes the common sense or Judgemental part.

18) Which AI technique enables the computers to understand the associations and relationships between objects and events?

- a. Heuristic Processing
- b. Cognitive Science
- c. Relative Symbolism
- d. Pattern Matching

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Answer: d. Pattern Matching

Explanation: Pattern matching is a way to check a given sequence of tokens in order to determine the presence of a given character or data in the given sequence. It allows computers to understand the relationship between objects and events.

19) The exploration problem is where_____.

- a. Agent contains the knowledge of State and actions.
- b. Agent does not contain the knowledge of State and actions.
- c. Only actions are known to the agent.
- d. None of the above

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Answer: b. Agent does not contain knowledge State and actions

Explanation: In Exploration problems, the agent does not contain the knowledge of state space and actions in advance. These are difficult problems and used in the real world.

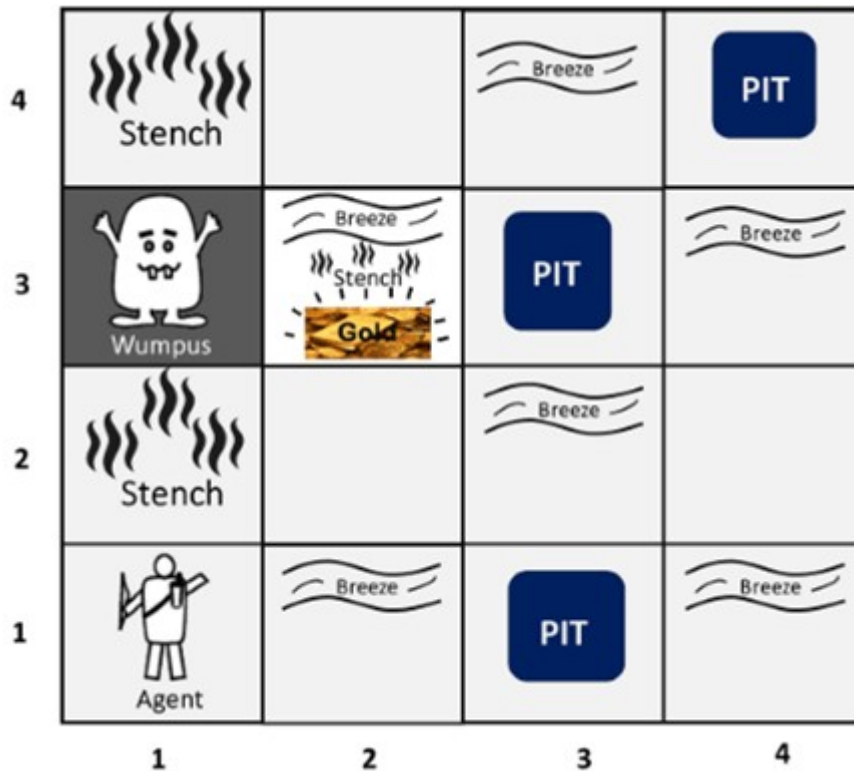
20) In the Wumpus World Problem, the reason for the uncertainty is that the agent's sensor gives only__

- a. Full & Global information
- b. Partial & Global Information
- c. Full & local information
- d. Partial & local Information

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Answer: d. Partial & local Information

Explanation: The Wumpus world is an example environment that is made of grids of squares surrounded by walls. Each square can have agents or objects. The world is used to demonstrate the worth of a knowledge-based agent and knowledge representation. In the environment, uncertainty arises as the agent can only perceive the close environment. The Wumpus world is represented in below image:



21) The search algorithm which is similar to the minimax search, but removes the branches that don't affect the final output is known as __.

- a. Depth-first search
- b. Breadth-first search
- c. Alpha-beta pruning
- d. None of the above

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Answer: c. Alpha-beta pruning

Explanation: Alpha-beta pruning algorithm is the modified version of the Minimax algorithm and returns the same moves as the original algorithm, but it removes all those nodes/branches that do not affect the final decision.

22) The maximum depth to which the alpha-beta pruning can be applied.

- a. Eight states
- b. Six states
- c. Ten states

- d. Any depth

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Answer: d. Any depth

Explanation: The Alpha-beta pruning can be applied to any depth of the tree and it can eliminate the entire subtree, if it is not affecting the final decision.

23) Among the given options, which is also known as inference rule?

- a. Reference
- b. Reform
- c. Resolution
- d. None of the above

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Answer: c. Resolution

Explanation: Resolution is also known as inference rule as it shows the complete inference rule when applied to any search algorithm.

24) Which of the following option is used to build complex sentences in knowledge representation?

- a. Symbols
- b. Connectives
- c. Quantifier
- d. None of the above

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Answer:

Explanation: Complex sentences are built by combining the atomic sentences using connectives.

25) Automatic Reasoning tool is used in_____.

- a. Personal Computers
- b. Microcomputers

- c. LISP Machines
- d. All of the above

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Answer: c. LISP Machine

Explanation: ART or Automatic Reasoning tool is used in LISP machines to understand the different aspects of reasoning.

26) If according to the hypothesis, the result should be positive, but in fact it is negative, then it is known as_____.

- a. False Negative Hypothesis
- b. False Positive Hypothesis
- c. Specialized Hypothesis
- d. Consistent Hypothesis

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Answer: b. False Positive Hypothesis

Explanation: The False Positive Hypothesis means that according to results, you have that condition, but in reality, you don't have it. Such as for a medical test, if someone is found Positive for a disease, but actually he doesn't have that disease, then it comes under the False Positive hypothesis.

27) A hybrid Bayesian Network consist_____.

- a. Discrete variables only
- b. Discontinuous Variable
- c. Both Discrete and Continuous variables
- d. Continuous Variable only

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Answer: c. Both Discrete and Continuous Variables

Explanation: The Hybrid Bayesian network contains both discrete and continuous variables as the numerical inputs. To define the hybrid network, both kinds of distributions are used at wide probability distribution.

28) The process of capturing the inference process as Single Inference Rule is known as:

- a. Clauses
- b. Ponens
- c. Generalized Modus Ponens
- d. Variables

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Answer: c. Generalized Modus Ponens

Explanation: For all inference process in FOL, the single inference rule can be used, which is called Generalized Modus Ponens. It is said to be the lifted version of Modus ponens.

Generalized Modus Ponens can be said as, " **P implies Q and P is asserted to be true, therefore Q must be True.**"

29) Which process makes two different Logical expressions look identical?

- a. Unification
- b. Lifting
- c. Inference Process
- d. None of the above

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Answer: a. Unification

Explanation: Unification is the process of making two different logical expressions identical by finding a substitution.

30) Which algorithm takes two sentences as input and returns a Unifier?

- a. Inference
- b. Hill-Climbing
- c. Unify algorithm
- d. Depth-first search

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Answer: c. Unify Algorithm

Explanation: The unify algorithm takes two atomic sentences and return a unifier. It is used for the unification process.

31) The PEAS in the task environment is about_____.

- a. Peer, Environment, Actuators, Sense
- b. Performance, Environment, Actuators, Sensors
- c. Perceiving, Environment, Actuators, Sensors
- d. None of the above

Hide Answer Workspace

Answer: b. Performance, Environment, Actuators, Sensors

Explanation: PEAS is a representation model on which an AI agent works. It is made up of four words:

- P: Performance
- E: Environment
- A: Actuators
- S: Sensors

32) In state-space, the set of actions for a given problem is expressed by the_____.

- a. Intermediate States
- b. Successor function that takes current action and returns next state
- c. Initial States
- d. None of the above

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Answer: b. Successor function that takes current action and returns next state

Explanation: The successor function provides a description of all possible actions and their next states, which means their outcomes.

33) In which search problem, to find the shortest path, each city must be visited once only?

- a. Map coloring Problem
- b. Depth-first search traversal on a given map represented as a graph
- c. Finding the shortest path between a source and a destination

- d. Travelling Salesman problem

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Answer: d. Travelling Salesman problem

Explanation: The TSP or Travelling Salesman problem is about finding the shortest possible route to visit each city only once and returning to the origin city when the list of all cities and distances between each pair of cities is given.

34) In the TSP problem of n cities, the time taken for traversing all cities, without having prior knowledge of the length of the minimum tour will be _____.

- a. $O(n)$
- b. $O(n^2)$
- c. $O(n!)$
- d. $O(n/2)$

Hide Answer Workspace

Answer: c. $O(n!)$

Explanation: In the TSP problem of n cities, the time taken for traversing all cities without having prior knowledge of the length of the minimum tour will be $O(n!)$.

35) Web Crawler is an example of _____.

- a. Intelligent Agent
- b. Problem-solving agent
- c. Simple reflex agent
- d. Model-based agent

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Answer: Intelligent Agent

Explanation: The web crawler is an example of Intelligent agents, which is responsible for collecting resources from the Web, such as HTML documents, images, text files, etc.

36) The main function of problem-solving agent is to _____.

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- a. Solve the given problem and reach the goal
- b. Find out which sequence of action will get it to the goal state.
- c. Both a & b
- d. None of the above

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Answer: Both a & b

Explanation: Problem-solving agents are the goal-based agents that use different search strategies and algorithms to solve a given problem.

37) In artificial Intelligence, knowledge can be represented as _____.

- i. Predicate Logic
- ii. Propositional Logic
- iii. Compound Logic
- iv. Machine Logic

- a. Both I and II
- b. Only II
- c. Both II and III
- d. Only IV

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Answer: a. Both I and II

Explanation: There are several techniques of [Knowledge representation in AI](#), and among them, one is Logical Representation. The logical representation can be done in two ways **Predicate Logic and Propositional Logic**, hence knowledge can be represented as both predicate and Propositional logic.

38) For propositional Logic, which statement is false?

- a. The sentences of Propositional logic can have answers other than True or False.
- b. Each sentence is a declarative sentence.
- c. Propositional logic is a knowledge representation technique in AI.
- d. None of the above

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Answer: a. The sentences of Propositional logic can have answers other than True or False

Explanation: Propositional Knowledge or PL is the simplest form of logic that is used to represent the knowledge, where all the sentences are propositions. In this, each sentence is a declarative sentence that can only be either true or False.

Such as, **It is Sunday today.** This sentence can be either true or false only.

39) First order logic Statements contains_____.

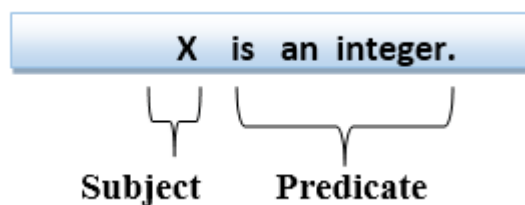
- a. Predicate and Preposition
- b. Subject and an Object
- c. Predicate and Subject
- d. None of the above

Hide Answer Workspace

Answer: c. Predicate and Subject

Explanation: The first-order logic is also known as the First-order predicate logic, which is another way of knowledge representation. The FOL statements contain two parts that are **subject and Predicate**.

For e.g., **X is an Integer**; In this, X is Subject and Is an Integer is Predicate.



40) A knowledge-based agent can be defined with _____ levels.

- a. 2 Levels
- b. 3 Levels
- c. 4 Levels
- d. None of the above

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Answer: b. 3 Levels

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Explanation: The knowledge-based agents have the capability of making decisions and reasoning to act efficiently. It can be viewed at three different levels, which are:

- **Knowledge Level**
 - **Logical Level**
 - **Implementation Level**
-

41) Ways to achieve AI in real-life are_____.

- a. Machine Learning
- b. Deep Learning
- c. Both a & b
- d. None of the above

Hide Answer Workspace

Answer: c. Both a &b

Explanation: Machine Learning and Deep Learning are the two ways to achieve AI in real life.

42) The main tasks of an AI agent are_____.

- a. Input and Output
- b. Moment and Humanly Actions
- c. Perceiving, thinking, and acting on the environment
- d. None of the above

Hide Answer Workspace

Answer: c. Perceiving, thinking, and acting on the environment

Explanation: The AI agent is the rational agent that runs in the cycle of Perceive, think, and act.

43) The probabilistic reasoning depends upon_____.

- a. Estimation
- b. Observations
- c. Likelihood

- d. All of the above

Hide Answer Workspace

Answer: d. All of the above

Explanation: The probabilistic reasoning is used to represent uncertain knowledge, where we are not sure about the predicates. It depends Upon Estimation, Observation, and likelihood of objects.

44) The inference engine works on _____.

- a. Forward Chaining
- b. Backward Chaining
- c. Both a and b
- d. None of the above

Hide Answer Workspace

Answer: c. Both a and b

Explanation: The inference engine is the component of the intelligent system in artificial intelligence, which applies logical rules to the knowledge base to infer new information from known facts. The first inference engine was part of the expert system. Inference engine commonly proceeds in two modes, which are:

- **Forward chaining**
 - **Backward chaining**
-

45) Which of the given statement is true for Conditional Probability?

- a. Conditional Probability gives 100% accurate results.
- b. Conditional Probability can be applied to a single event.
- c. Conditional Probability has no effect or relevance on independent events.
- d. None of the above.

Hide Answer Workspace

Answer: c. Conditional Probability has no effect or relevance on independent events.

Explanation: The conditional probability is said as the probability of occurring an event when another event has already occurred. And Independent events are those that are not affected by the occurrence of other events; hence conditional probability has no effect or relevance on independent events.

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46) After applying conditional Probability to a given problem, we get_____

- a. 100% accurate result
- b. Estimated Values
- c. Wrong Values
- d. None of the above

Hide Answer Workspace

Answer: b. Estimated Values

Explanation: Like all probability theories and methods, Conditional Probability also provides the estimated result value, which means the probability of an event to occur, not a 100% accurate result.

47) The best AI agent is one which_____

- a. Needs user inputs for solving any problem
- b. Can solve a problem on its own without any human intervention
- c. Need a similar exemplary problem in its knowledge base
- d. All of the above

Hide Answer Workspace

Answer: b. Can solve a problem on its own without any human intervention

Explanation: The best AI agent is one that can solve the problem on its own without any human intervention.

48) The Bayesian Network gives_____

- a. A complete description of the problem
- b. Partial Description of the domain
- c. A complete description of the domain
- d. None of the above

Hide Answer Workspace

Answer: c. A complete description of the domain

Explanation: A [Bayesian network](#) is a probabilistic graphical model that represents a set of variables and their conditional dependencies using a directed acyclic graph. It gives a complete description of the domain.

49) In LISP, the addition of 5+8 is entered as_____.

- a. 5+8
- b. 5 add 8
- c. 5+8=
- d. (+5 8)

Hide Answer Workspace

Answer: d. (+5 8)

Explanation: The sum of two variables a & b can be entered as (+a b). Hence the sum of 5 and 8 can be entered as (+5 8).

50) An Algorithm is said as Complete algorithm if_____

- a. It ends with a solution (if any exists).
- b. It begins with a solution.
- c. It does not end with a solution.
- d. It contains a loop

Hide Answer Workspace

Answer: a. It ends with a solution (if any exists).

Explanation: An algorithm is only said the complete algorithm if it ends with a solution (if it exists).

51) Which statement is valid for the Heuristic function?

- a. The heuristic function is used to solve mathematical problems.
- b. The heuristic function takes parameters of type string and returns an integer value.
- c. The heuristic function does not have any return type.
- d. The heuristic function calculates the cost of an optimal path between the pair of states.

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Answer: d. The heuristic function calculates the cost of an optimal path between the pair of states

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Explanation: The heuristic function is used in [Informed search in AI](#) to find the most promising path in the search. It estimates the closeness of the current state and calculates the cost of an optimal path between the pair of states. It is represented by $h(n)$.

52) Which of the given element improve the performance of AI agent so that it can make better decisions?

- a. Changing Element
- b. Performance Element
- c. Learning Element
- d. None of the above

Hide Answer Workspace

Answer: c. Learning Element

Explanation: The learning element improves the performance of an AI agent while solving a given problem, so that it can make better decisions.

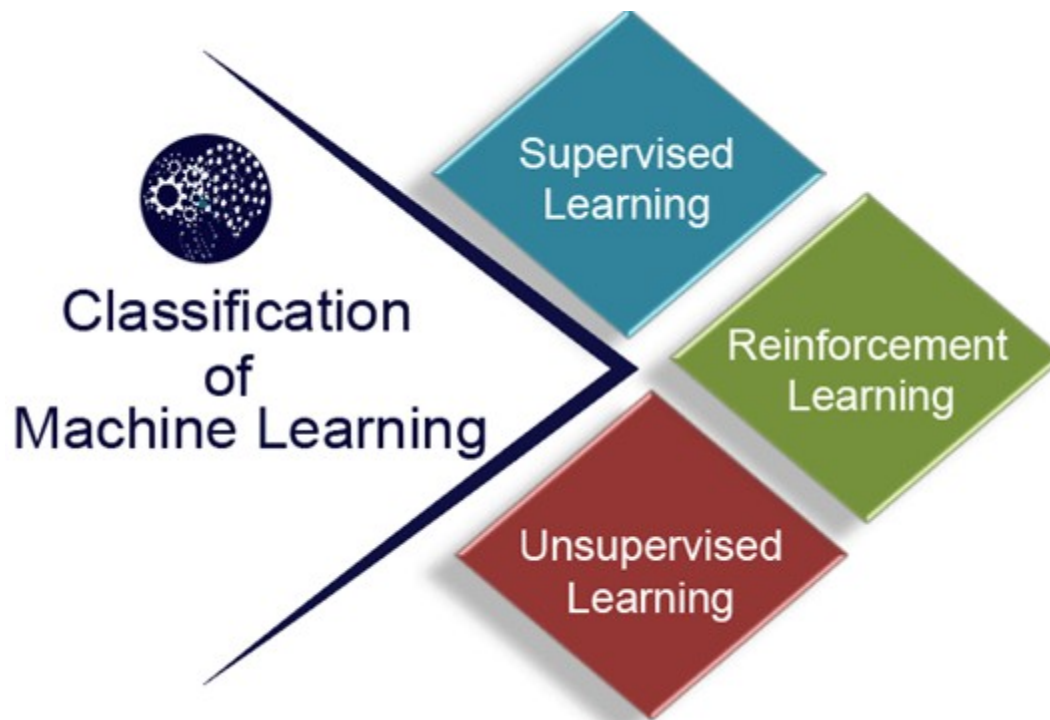
53) How many types of Machine Learning are there?

- a. 1
- b. 2
- c. 3
- d. 4

Hide Answer Workspace

Answer: c. 3

Explanation: There are three types of Machine Learning techniques, which are Supervised Learning, Unsupervised Learning, and Reinforcement Learning.



54) The decision tree algorithm reaches its destination using_____.

- a. Single Test
- b. Two Test
- c. Sequence of test
- d. No test

Hide Answer Workspace

Answer: c. Sequence of test

Explanation: A decision tree is the supervised machine learning technique that can be used for both Classification and Regression problems. It reaches its destination using a Sequence of Tests.

55) In LISP programming, the square root is entered as_____.

- a. Sqrt(x)
- b. (sqrt x)
- c. x/2
- d. none of the above

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Answer: (sqrt x)

Explanation: In LISP programming, the square root of any variable x is entered as (sqrt x).

1. Who is the father of Artificial Intelligence?

- A. Doug Cutting
- B. John McCarthy
- C. William S.
- D. Rasmus Lerdorf

[View Answer](#)

Ans : B

Explanation: the father of Artificial Intelligence is John McCarthy.

2. Artificial Intelligence is a way of _____.

- A. making a computer
- B. a computer-controlled robot
- C. software think intelligently
- D. All of the above

[View Answer](#)

Ans : D

Explanation: Artificial Intelligence is a way of making a computer, a computer-controlled robot, or a software think intelligently, in the similar manner the intelligent humans think.

3. What are the main goals of AI?

- A. To Create Expert Systems
- B. To Implement Human Intelligence in Machines
- C. Both A and B
- D. None of the Above

[View Answer](#)

Ans : C

Explanation: the main goals of AI is to Create Expert Systems and to Implement Human Intelligence in Machines

4. A major thrust of AI is in the development of computer functions associated with human intelligence.

- A. TRUE
- B. FALSE
- C. AI is not associated with human intelligence
- D. None of the Above

[View Answer](#)

Ans : A

Explanation: A major thrust of AI is in the development of computer functions associated with human intelligence, such as reasoning, learning, and problem solving.

5. Which of the following areas can contribute to build an intelligent system?

- A. Philosophy
- B. Biology
- C. Sociology
- D. All of the above

[View Answer](#)

Ans : D

Explanation: Philosophy, Sociology, Biology all three areas can contribute to build an intelligent system.

6. Which of the following areas can not contribute to build an intelligent system?

- A. Neuron science
- B. Maths
- C. Computer Science
- D. Geology

[View Answer](#)

Ans : D

Explanation: Geology can not contribute to build an intelligent system.

7. A computer program with AI can not answer the generic questions it is meant to solve.

- A. TRUE
- B. FALSE
- C. AI is not used to answer question
- D. None of the Above

[View Answer](#)

Ans : B

Explanation: A computer program with AI can answer the generic questions it is meant to solve.

8. How many types of AI is there?

- A. 2
- B. 3
- C. 4
- D. 5

[View Answer](#)

Ans : C

Explanation: There are four types of artificial intelligence

9. Which of the following is not the type of AI?

- A. Reactive machines
- B. Unlimited memory
- C. Theory of mind
- D. Self-awareness

View Answer

Ans : B

Explanation: There are four types of artificial intelligence: reactive machines, limited memory, theory of mind and self-awareness.

10. AI Technique is a manner to organize and use the knowledge efficiently in such a way that _____.

- A. It should be perceivable by the people who provide it.
- B. It should be easily modifiable to correct errors.
- C. Both A and B
- D. None of the Above

View Answer

Ans : B

Explanation: Both A and B is Correct

11. Which of the following is an application of AI?

- A. Gaming
- B. Expert Systems
- C. Vision Systems
- D. All of the above

View Answer

Ans : D

Explanation: All options are correct

12. In which year John McCarthy coined the term Artificial Intelligence?

- A. 1950
- B. 1953
- C. 1956
- D. 1959

View Answer

Ans : C

Explanation: In 1956, John McCarthy coined the term Artificial Intelligence.

13. In which university the first demonstration of AI program run?

- A. Carnegie Mellon University.
- B. Oxford University
- C. Cambridge University
- D. Stanford University

View Answer

Ans : A

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Explanation: John McCarthy coined the term Artificial Intelligence. Demonstration of the first running AI program at Carnegie Mellon University.

14. Which of the following is not an application of AI?

- A. Intelligent Robots
- B. Handwriting Recognition
- C. Speech Recognition
- D. Content mining

[View Answer](#)

Ans : D

Explanation: Content mining is not an application of AI.

15. Expert Ease was developed under the direction of _____

- A. John McCarthy
- B. Donald Michie
- C. Lofti Zadeh
- D. Alan Turing

[View Answer](#)

Ans : B

Explanation: Expert Ease was developed under the direction of Donald Michie

16. The traditional way to exit and LISP system is to enter _____

- A. quit
- B. exit
- C. bye
- D. ok

[View Answer](#)

Ans : B

Explanation: The traditional way to exit and LISP system is to enter exit.

17. What is Artificial intelligence?

- A. Putting your intelligence into Computer
- B. Programming with your own intelligence
- C. Making a Machine intelligent
- D. Playing a Game

[View Answer](#)

Ans : C

Explanation: Because AI is to make things work automatically through machine without using human effort. Machine will give the result with just giving input from human. That means the system or machine will act as per the requirement.

18. When a top-level function is entered, the LISP processor do(es)?

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- A. It reads the function entered
- B. It evaluates the function and the function's operands
- C. It prints the results returned by the function
- D. All of the above

View Answer

Ans : D

Explanation: All options are correct

19. Which of the following, is a component of an expert system?

- A. inference engine
- B. knowledge base
- C. user interface
- D. All of the above

View Answer

Ans : D

Explanation: All options are Correct.

20. A process that is repeated, evaluated, and refined is called _____

- A. diagnostic
- B. descriptive
- C. interpretive
- D. iterative

View Answer

Ans : D

Explanation: A process that is repeated, evaluated, and refined is called iterative.

1.

The “Father of Artificial Intelligence” is:

Alan Turing

Charles Babbage

John McCarthy

None of the above

Hide

Wrong Answer

John McCarthy is also known as the “Father of Artificial Intelligence”.

2.

Blind Search can be used for which of the following situations?

Real-Life Simulation

Small Search Space

Advanced Game Theory

None of the above

Hide

Wrong Answer

Blind Search doesn't contain any domain information such as closeness and hence it is best used for a Small Search Space.

3.

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In how many category processes is Artificial Intelligence classified in?

Depends on the input nature

5

2

3

Hide

Wrong Answer

AI is classified into 3 categories processes: Sensing, Reasoning, and Acting.

4.

If a machine can change its course of action based on the external environment on its own, the machine is called?

Mobile

Intelligent

Both A and B

None of the above

Hide

Wrong Answer

Machines that can change their course of action by making their own decisions are said to be Intelligent.

5.

Which of the following is the common language for Artificial Intelligence?

Python

Java

Lisp

PHP

Hide

Wrong Answer

While programming can be done in any language, in today's world Python has become the go-to language for AI and ML-related tasks due to its vast and diverse library functionalities.

6.

Which of the following is not an application of artificial intelligence?

Computer Vision

Natural Language Processing

Database Management System

Digital Assistants

Hide

Wrong Answer

Other than Database Management System, all the other options are viable fields of study in Artificial Intelligence.

7.

The proposition symbols in AI are?

true and false

true, false, and null

true

false

Hide

Wrong Answer

true and false are the 2 proposition symbols in AI.

8.

Which of the following symbols in AI are logical symbols?

Negation

Conjunction

Implication

All of the above

Hide

Wrong Answer

The logical symbols in AI are: Negation, Conjunction, Disjunction, Implication, and Biconditional symbols.

9.

Which of the following are informed search methods?

Best First Search

A* Search

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Memory Bound Heuristic Search

All of the above

Hide

Wrong Answer

All of the above options are informed search methods used in Artificial Intelligence.

10.

The name of the Artificial Intelligence system developed by Daniel Bobrow was?

BACON

SIMD

STUDENT

None of the above

Hide

Wrong Answer

The name of the Artificial Intelligence system developed by Daniel Bobrow was STUDENT developed in the Lisp Language in 1964.

11.

Which of the following is a type of artificial intelligence agent?

Learning AI Agent

Simple Reflex AI Agent

Goal-Based AI Agent

All of the above

Hide

Wrong Answer

There are a total of 5 AI agents: Simple Reflex Agent, Model-Based Reflex Agent, Goal-based reflex agent, utility-based agent, learning agent.

12.

How many types of recognition are there in AI?

5

3

2

1

Hide

Wrong Answer

There are 3 types of recognition in AI: Biometric Identification, Content-Based Image Retrieval, Handwriting Recognition.

13.

Procedural Domain Knowledge in a rule-based system, is in the form of?

Meta-Rules

Control Rules

Production Rules

None of the above

Hide

Wrong Answer

Procedural Domain Knowledge in a rule-based system, is in the form of Production Rules.

14.

What are the different types of Artificial Intelligence approaches?

Strong Approach

Weak Approach

Applied Approach

All of the above

Hide

Wrong Answer

All the given options are different types of AI approaches used in different scenarios.

15.

‘The Imitation Game’ was the original name of?

LISP

The Turing Test

The Halting Problem

None of the above

Hide

Wrong Answer

The Turing Test was originally called the 'The Imitation Game' by its creator.

16.

Which of the following architecture is also known as systolic arrays?

MISD

SISD

SIMD

None of the above

Hide

Wrong Answer

MISD architecture is also known as systolic arrays.

17.

Decisions of Victory/Defeat are made in Game trees using which algorithm?

DFS

BFS

Heuristic Search

Min/Max Algorithm

Hide

Wrong Answer

The Min/Max algorithm is preferred over other search-based algorithms in Game trees to make win/loss decisions optimally.

18.

The correct ways to solve a problem of state-space search are?

Forward from the initial state

Backward from the goal

Both A and B

None of the above

Hide

Wrong Answer

The only 2 ways to solve a problem of state-space search problem are given in options (A) and (B).

19.

Machines running LISP are also called?

AI Workstations

Time-Sharing Terminals

Both A and B

None of the above

Hide

Wrong Answer

Machines running LISP are also called AI workstations.

20.

How does an AI agent interacts with its environment?

Using sensors and perceivers

Using only sensors

Using only perceivers

None of the above

Hide

Wrong Answer

An AI agent perceives and acts upon the environment using Sensors and Actuators. With Sensors, it senses the surrounding, and with Actuators, it acts on it.

21.

What is the work of Task Environment and Rational Agents?

Problem and Solution

Solution and Problem

Observation and Problem

Observation and Solution

Hide

Wrong Answer

Task Environments provides an issue and Rational Agents provides the corresponding solution.

22.

How many types of observing environments are there?

2

3

1

0

Hide

Wrong Answer

There are 2 types of observing environments: Fully and Partial.

23.

How many types of quantification are there in AI?

1

2

3

4

Hide

Wrong Answer

There are 2 types of quantification in AI: Universal and Existential.

24.

Out of the given options, which of the following algorithms uses the least memory?

DFS

BFS

Both A and B are the same

Cannot be compared

Hide

Wrong Answer

DFS uses the lesser amount of memory because at a time the path from the node to a root is only stored in the memory stack unlike BFS which will have to store the whole tree within it.

25.

Machines that try to imitate human intuition while handling vague information lie in the field of AI called?

Functional Logic

Fuzzy Logic

Boolean Logic

Human Logic

Hide

Wrong Answer

Fuzzy Logic allows machines to mimic human intuition when provided with information that is vague in nature.

26.

What is meant by a "Complete Algorithm"?

If a solution exists, the algorithm will find it before terminating.

It will find the solution in a finite amount of time.

Both A and B.

None of the above.

Hide

Wrong Answer

A complete algorithm is an algorithm that finds a solution to a problem(if it exists) in finite time.

27.

The measure of performance of an AI agent is measured using?

Learning Agent.

Changing Agent.

Both A and B.

None of the above.

Hide

Wrong Answer

The performance of an AI agent is improved using the learning agent.

28.

Which of the following are valid Machine Learning algorithms?

Linear Regression

K Means Clustering

Naive Bayes

All of the above

Hide

Department of software engineering

Wrong Answer

All of the above algorithms given above are valid ML algorithms.

29.

What are different machine learning methods?

Memorization

Analogy

Deduction

All of the above

Hide

Wrong Answer

The machine learning methods include Memorization, Analogy, and Deduction.

30.

Exploratory Learning is another name for?

Supervised learning

Unsupervised learning

Reinforcement learning

None of the above

Hide

Wrong Answer

In unsupervised learning, no external supervision is required so the machine learns on the basis of exploring the datasets themselves.

31.

The things considered in the design of a learning element are?

Components

Feedback

Representation

All of the above

Hide

Wrong Answer

The 3 main things to be considered in the design of a learning element are: Components, Feedback, and Representation.

32.

The different types of machine learning are?

Supervised

Unsupervised

Reinforcement

All of the above

Hide

Wrong Answer

The 3 types of machine learning are: Supervised, Unsupervised, and Reinforcement Learning.

33.

How is a decision reached upon by a decision tree?

No test

Single Test

Double Test

Multiple sequences of tests

Hide

Wrong Answer

A decision tree makes its decision based on the results of a sequence of tests.

34.

A hybrid Bayesian Network consists of?

Discrete Variables

Continuous Variables

Both A and B

None of the above

Hide

Wrong Answer

Discrete and Continuous variables both act as numerical inputs to a hybrid Bayesian network.

35.

AI agents are composed of?

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Architecture

Program

Both A and B

None of the above

Hide

Wrong Answer

AI agents implement mapping for perceptions to actions.

36.

For external action selection, which element is used in the agent?

Perceive

Performance

Actuator

None of the above

Hide

Wrong Answer

The performance element is used for external action selection in AI agents.

37.

How is the brightness of a pixel on the screen increased?

Increasing the amount of light directed on the screen

Using mirrors

Using sound waves

None of the above

Hide

Wrong Answer

The brightness of a pixel is directly proportional to the amount of light incident on the screen.

38.

The process of making 2 logical expressions look identical is called?

Lifting

Unification

Both A and B

None of the above

Hide

Wrong Answer

Unification is the process of making 2 logical expressions look identical.

39.

PEAS is an abbreviation for?

Peace, Environment, Action, Sense

Peer, Environment, Actuators, Sensors

Performance, Environment, Actuators, Sensors

Performance, Environment, Actuators, Sense

Hide

Wrong Answer

PEAS is an abbreviation for Performance, Environment, Actuators, Sensors.

40.

Knowledge in AI can be represented as?

Predicate Logic

Propositional Logic

Both A and B

None of the above

Hide

Wrong Answer

The representation of knowledge in AI can be done both as Predicate and Propositional Logic.

41.

Which of the following are appropriate levels for a knowledge-based AI agent?

Knowledge Level

Logical Level

Implementation Level

All of the above

Hide

Wrong Answer

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The 3 main levels for a knowledge-based AI agent are: Knowledge, Logical, and Implementation level.

42.

How can we achieve Artificial Intelligence in real life?

Machine Learning

Deep Learning

Both A and B

None of the above

Hide

Wrong Answer

Machine learning and Deep Learning are 2 of the ways to implement Artificial Intelligence in real life.

43.

Which of the following are heuristic search algorithms?

Best First Search Algorithm

A* Search Algorithm

Both A and B

None of the above

Check Answer

44.

How do we multiply 2 numbers in LISP?

* a b

a * b

a mul b

None of the above

Hide

Wrong Answer

The syntax to multiply 2 numbers in LISP is correctly shown in option A.

45.

Inference engines work on the principle of?

Backward Chaining

Forward Chaining

Both A and B

None of the above

Hide

Wrong Answer

The modes on which inference engines work are forward and backward chaining.

46.

The process of breaking an image into parts is called?

Smoothing

Segmentation

Processing

Break Down

Hide

Wrong Answer

Segmentation is the process of breaking an image into groups based on the similarity of pixels.

47.

Which of the following are valid 3D image processing techniques?

Motion

Texture

Contour

All of the above

Hide

Wrong Answer

All of the above options are valid 3D image processing techniques.

48.

Parsing is used for?

Parse tree buildup

Data Interpretation

Sound Recognition

None of the above

Hide

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Wrong Answer

The process of building up a parse tree for an input string is called parsing.

49.

Artificial Intelligence is associated with computers of which generation?

Second

First

Fifth

Third

Hide

Wrong Answer

Fifth Generation computers are said to be able to have Artificial Intelligence within them.

50.

Components of an expert system are?

Knowledge base

User interface

Inference engine

All of the above

Hide

Wrong Answer

All the above options comprise the components of an expert system.

51.

Which of the following search method takes less memory space?

Depth First Search

Breadth-First Search

Linear Search

Optimal Search

Hide

Wrong Answer

Depth First Search takes the least memory consumption because only the nodes on the current path are stored.

52.

Which of the following is not a type of AI?

Weak AI

Theory of Mind

Reactive Machines

All of the above

Hide

Wrong Answer

Reactive Machines are not a type of AI based on functionality.

53.

What is state space in AI?

Collection of all the problem states.

A specific problem states out of all the problem states.

Both A and B.

None of the above

Hide

Wrong Answer

State Space in AI is defined as the collection of all the problem states.